

Wayside report

Generated: 2026-09-04T00:00:00+00:00

Summary

Analysis of capture `4sics-slice.pcap` raised 5 findings requiring attention.

Scope

The capture carries 40 packets. Protocol(s) recognised in this capture: modbus-tcp, recognised by the shape of the segment payload, never by port number. Traffic whose protocol was not recognised has a row in the communication matrix labelled `tcp` and is not the basis of any finding.

Capture time window: from 1445499126.04817 to 1445499126.8009 (Unix epoch timestamps). Snaplen read from the capture header: 65535 bytes. Frames truncated by snaplen: 0. Events recognised with low confidence: 0.

Methodology

Every finding carries an indicator of behaviour observed in network traffic, never a judgement on whether an installation does or does not meet the requirements of a standard. Finding severity follows the documented rubric criteria below (version 1.0), not an invented scale:

- low: An observation with limited security impact and no direct path to disrupting the operation of the process.
- medium: A departure from good practice which, combined with another condition, may lead to disrupting the operation of the process.
- high: An operation that by itself allows the state of the process to be influenced without authentication or authorisation of the sender.
- critical: A condition enabling immediate and direct interference with the safety of the process, with no further conditions required.

Asset inventory

10.10.10.20

- ip: 10.10.10.20 (observed)
- mac: 00:1c:06:27:64:11 (observed)
- Vendor: Siemens Numerical Control Ltd., Nanjing (inferred:oui-lookup)
- Unit ID sub-addresses: not determined (not-derivable-passively)
- Gateway: not determined (not-derivable-passively)
- Role: undetermined (not-derivable-passively)
- Role evidence: Zero Modbus events associated with this address in this capture (requests sent: 0, requests received: 0). (observed)
- Role confidence: low (inferred:event-count-and-direction)

10.10.10.10

- ip: 10.10.10.10 (observed)
- mac: 28:63:36:89:59:82 (observed)
- Vendor: Siemens AG (inferred:oui-lookup)
- Unit ID sub-addresses: not determined (not-derivable-passively)
- Gateway: not determined (not-derivable-passively)
- Role: undetermined (not-derivable-passively)
- Role evidence: Zero Modbus events associated with this address in this capture (requests sent: 0, requests received: 0). (observed)
- Role confidence: low (inferred:event-count-and-direction)

192.168.88.50

- ip: 192.168.88.50 (observed)
- mac: 00:05:e4:01:24:d3 (observed)
- Vendor: Red Lion Controls Inc. (inferred:oui-lookup)
- Unit ID sub-addresses: 1 (observed)
- Gateway: not determined (not-derivable-passively)
- Role: Modbus server (inferred:modbus-traffic-direction)
- Role evidence: Modbus requests sent by this address: 0; Modbus requests received by this address: 1. (observed)
- Role confidence: low (inferred:event-count-and-direction)

192.168.2.44

- ip: 192.168.2.44 (observed)
- mac: 00:07:7c:1a:61:83 (observed)
- Vendor: Westermo Network Technologies AB (inferred:oui-lookup)
- Unit ID sub-addresses: not determined (not-derivable-passively)
- Gateway: not determined (not-derivable-passively)
- Role: Modbus client (inferred:modbus-traffic-direction)
- Role evidence: Modbus requests sent by this address: 5; Modbus requests received by this address: 0. (observed)
- Role confidence: medium (inferred:event-count-and-direction)

192.168.88.100

- ip: 192.168.88.100 (observed)
- mac: 00:e0:62:40:57:66 (observed)
- Vendor: HOST ENGINEERING (inferred:oui-lookup)
- Unit ID sub-addresses: 1 (observed)
- Gateway: not determined (not-derivable-passively)
- Role: Modbus server (inferred:modbus-traffic-direction)
- Role evidence: Modbus requests sent by this address: 0; Modbus requests received by this address: 1. (observed)
- Role confidence: low (inferred:event-count-and-direction)

192.168.88.20

- ip: 192.168.88.20 (observed)
- mac: 00:a0:45:6f:4b:83 (observed)
- Vendor: Phoenix Contact GmbH & Co. KG (inferred:oui-lookup)
- Unit ID sub-addresses: 1 (observed)
- Gateway: not determined (not-derivable-passively)
- Role: Modbus server (inferred:modbus-traffic-direction)
- Role evidence: Modbus requests sent by this address: 0; Modbus requests received by this address: 1. (observed)
- Role confidence: low (inferred:event-count-and-direction)

192.168.88.60

- ip: 192.168.88.60 (observed)
- mac: 00:90:e8:26:40:23 (observed)
- Vendor: MOXA TECHNOLOGIES CORP., LTD. (inferred:oui-lookup)
- Unit ID sub-addresses: 1 (observed)
- Gateway: not determined (not-derivable-passively)
- Role: Modbus server (inferred:modbus-traffic-direction)
- Role evidence: Modbus requests sent by this address: 0; Modbus requests received by this address: 1. (observed)

- Role confidence: low (inferred:event-count-and-direction)

192.168.88.61

- ip: 192.168.88.61 (observed)

- mac: 00:90:e8:27:8c:37 (observed)

- Vendor: MOXA TECHNOLOGIES CORP., LTD. (inferred:oui-lookup)

- Unit ID sub-addresses: 1 (observed)

- Gateway: not determined (not-derivable-passively)

- Role: Modbus server (inferred:modbus-traffic-direction)

- Role evidence: Modbus requests sent by this address: 0; Modbus requests received by this address: 1. (observed)

- Role confidence: low (inferred:event-count-and-direction)

Communication matrix

Session: 0

Source: 10.10.10.20:49156

Target: 10.10.10.10:102

Direction: 10.10.10.20:49156 -> 10.10.10.10:102 (inferred:first-observed-sender)

Protocol: tcp

Volume (B): 634

Packets: 6

Initiating party: not determined (not-derivable-passively)

Session: 1

Source: 192.168.2.44:58597

Target: 192.168.88.50:502

Direction: 192.168.2.44:58597 -> 192.168.88.50:502 (observed)

Protocol: modbus-tcp

Volume (B): 260

Packets: 4

Initiating party: 192.168.2.44:58597 (observed)

Session: 2

Source: 192.168.2.44:58601

Target: 192.168.88.100:502

Direction: 192.168.2.44:58601 -> 192.168.88.100:502 (observed)

Protocol: modbus-tcp

Volume (B): 320

Packets: 5

Initiating party: 192.168.2.44:58601 (observed)

Session: 3

Source: 192.168.2.44:58599

Target: 192.168.88.20:502

Direction: 192.168.2.44:58599 -> 192.168.88.20:502 (observed)

Protocol: modbus-tcp

Volume (B): 260

Packets: 4

Initiating party: 192.168.2.44:58599 (observed)

Session: 4

Source: 192.168.2.44:58598

Target: 192.168.88.60:44818

Direction: 192.168.2.44:58598 -> 192.168.88.60:44818 (observed)

Protocol: tcp

Volume (B): 308

Packets: 4

Initiating party: 192.168.2.44:58598 (observed)

Session: 5

Source: 192.168.2.44:58600

Target: 192.168.88.60:502

Direction: 192.168.2.44:58600 -> 192.168.88.60:502 (observed)

Protocol: modbus-tcp

Volume (B): 367

Packets: 5

Initiating party: 192.168.2.44:58600 (observed)

Session: 6

Source: 192.168.2.44:58602

Target: 192.168.88.61:502

Direction: 192.168.2.44:58602 -> 192.168.88.61:502 (observed)

Protocol: modbus-tcp

Volume (B): 433

Packets: 6

Initiating party: 192.168.2.44:58602 (observed)

Limitations

- Scope of this run: addresses observed 8, sessions with payload 7, sessions without a single segment carrying payload 2, the capture time window is 0.75273 s long.
- This report describes only traffic that reached the capture point. A device absent from the result is not a device absent from the network - it is a device whose traffic did not pass this point.
- A device sitting behind a protocol gateway is visible only under the address of that gateway. A network address in this report may therefore correspond to more than one physical device.
- Multiple hosts hidden behind a single address after address translation are indistinguishable from this point. One inventory row may correspond to more than one device.
- TCP sessions made up exclusively of packets without payload: 2. They have no row in the communication matrix, because they carry no segment to recognise - they are counted here so they do not vanish without a trace.

This report comes from a vertical slice: one capture, one check, one standard clause. The zone and conduit model is a single-zone placeholder derived automatically from this capture, not a designed network topology. Standard clause numbering is provisional and awaits collation against a legal copy of the standard.

Fields that cannot be established from this capture, grouped by field name:

- section `assets`, field `gateway`: 8 of 8 entries
- section `assets`, field `role`: 2 of 8 entries
- section `assets`, field `unit\_ids`: 3 of 8 entries
- section `comm\_matrix`, field `initiator`: 1 of 7 entries

Findings

Use of an industrial protocol without an authentication mechanism in the observed communication

- Check identifier: unauthenticated-industrial-protocol
- Severity: high (risk: serious)

- Session parties: 192.168.2.44:58597 -> 192.168.88.50:502
- Evidence: packet no. 29, session no. 1
- Rationale: This finding concerns the mere use of a protocol that has no mechanism for authenticating the sender, regardless of whether a write operation occurred in this capture. Any host that can see this network segment may send a command the device will execute, and a read observed in the capture proves only that the communication path exists and is open. This is a property of the protocol, not a decision or an omission of the installation operator - in an older industrial installation there is often no alternative.
- Standard citation: IEC-62443-3-3 SR 1.2
 - Clause scope (own description, not a title from the copy): Software process and device identification and authentication
 - Paraphrase: The clause concerns ensuring that every software process and every device connecting to the control system is uniquely identified and authenticated - as distinct from human users, who are covered by a separate clause of this catalogue.
 - Status: PROVISIONAL, UNVERIFIED (The clause numbering and the paraphrase text await collation against a legal copy of IEC 62443-3-3 in phase 4. Until then the entry is provisional and does not constitute a confirmed citation.)
- Standard citation: CLC/TS 50701 undetermined-1
 - Clause scope (own description, not a title from the copy): Security requirements for railway signalling and control systems
 - Paraphrase: The document concerns cybersecurity requirements placed on railway traffic control and signalling systems, including protection of their correct operation against deliberate and accidental compromise.
 - Status: PROVISIONAL, UNVERIFIED (The document has the status of a CENELEC technical specification, not a European standard, so it is applied voluntarily - the report does not present it as a mandatory basis (decision 0004). The clause numbering has not been collated against a copy, because the project does not buy that copy; the entry is provisional for that reason (decision 0006).)
- Remediation: Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer.

Use of an industrial protocol without an authentication mechanism in the observed communication

- Check identifier: unauthenticated-industrial-protocol
- Severity: high (risk: serious)
- Session parties: 192.168.2.44:58601 -> 192.168.88.100:502
- Evidence: packet no. 34, session no. 2
- Rationale: This finding concerns the mere use of a protocol that has no mechanism for authenticating the sender, regardless of whether a write operation occurred in this capture. Any host that can see this network segment may send a command the device will execute, and a read observed in the capture proves only that the communication path exists and is open. This is a property of the protocol, not a decision or an omission of the installation operator - in an older industrial installation there is often no alternative.
- Standard citation: IEC-62443-3-3 SR 1.2
 - Clause scope (own description, not a title from the copy): Software process and device identification and authentication
 - Paraphrase: The clause concerns ensuring that every software process and every device connecting to the control system is uniquely identified and authenticated - as distinct from human users, who are covered by a separate clause of this catalogue.
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against a legal copy of IEC 62443-3-3 in phase 4. Until then the entry is provisional and does not constitute a confirmed citation.)

- Standard citation: CLC/TS 50701 undetermined-1

- Clause scope (own description, not a title from the copy): Security requirements for railway signalling and control systems

- Paraphrase: The document concerns cybersecurity requirements placed on railway traffic control and signalling systems, including protection of their correct operation against deliberate and accidental compromise.

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- Remediation: Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer.

Use of an industrial protocol without an authentication mechanism in the observed communication

- Check identifier: unauthenticated-industrial-protocol

- Severity: high (risk: serious)

- Session parties: 192.168.2.44:58599 -> 192.168.88.20:502

- Evidence: packet no. 32, session no. 3

- Rationale: This finding concerns the mere use of a protocol that has no mechanism for authenticating the sender, regardless of whether a write operation occurred in this capture. Any host that can see this network segment may send a command the device will execute, and a read observed in the capture proves only that the communication path exists and is open. This is a property of the protocol, not a decision or an omission of the installation operator - in an older industrial installation there is often no alternative.

- Standard citation: IEC-62443-3-3 SR 1.2

- Clause scope (own description, not a title from the copy): Software process and device identification and authentication

- Paraphrase: The clause concerns ensuring that every software process and every device connecting to the control system is uniquely identified and authenticated - as distinct from human users, who are covered by a separate clause of this catalogue.

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- Paraphrase: The document concerns cybersecurity requirements placed on railway traffic control and signalling systems, including protection of their correct operation against deliberate and accidental compromise.

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- Remediation: Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer.

Use of an industrial protocol without an authentication mechanism in the observed communication

- Check identifier: unauthenticated-industrial-protocol
- Severity: high (risk: serious)
- Session parties: 192.168.2.44:58600 -> 192.168.88.60:502
- Evidence: packet no. 33, session no. 5
- Rationale: This finding concerns the mere use of a protocol that has no mechanism for authenticating the sender, regardless of whether a write operation occurred in this capture. Any host that can see this network segment may send a command the device will execute, and a read observed in the capture proves only that the communication path exists and is open. This is a property of the protocol, not a decision or an omission of the installation operator - in an older industrial installation there is often no alternative.
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- Remediation: Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer.

Use of an industrial protocol without an authentication mechanism in the observed communication

- Check identifier: unauthenticated-industrial-protocol
- Severity: high (risk: serious)
- Session parties: 192.168.2.44:58602 -> 192.168.88.61:502
- Evidence: packet no. 35, session no. 6
- Rationale: This finding concerns the mere use of a protocol that has no mechanism for authenticating the sender, regardless of whether a write operation occurred in this capture. Any host that can see this network segment may send a command the device will execute, and a read observed in the capture proves only that the communication path exists and is open. This is a property of the protocol, not a decision or an omission of the installation operator - in an older industrial installation there is often no alternative.
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- Paraphrase: The clause concerns ensuring that every software process and every device connecting to the control system is uniquely identified and authenticated - as distinct from human users, who are covered by a separate clause of this catalogue.
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- Remediation: Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer.

Recommendations

- Restrict at the network level the set of hosts that may open a session to the controller at all, through segmentation and access control lists. The protocol itself cannot be authenticated without replacing devices or without an intermediary layer. (applies to 5 findings)